torch.nn.utils.weight_norm.weight_norm

https://pytorch.org/docs/stable/generated/torch.nn.utils.weight_norm.weight_

Description:

Apply weight normalization to a parameter in the given module. Weight normalization is a reparameterization that decouples the magnitude of a weight tensor from its direction. This replaces the parameter specified by name (e.g. 'weight') with two parameters: one specifying the magnitude (e.g. 'weight_g') and one specifying the direction (e.g. 'weight_v'). Weight normalization is implemented via a hook that recomputes the weight tensor from the magnitude and direction before every forward() call. By default, with dim=0, the norm is computed independently per output channel/plane. To compute a norm over the entire weight tensor, use dim=None. See https://arxiv.org/abs/1602.07868

Function Signature

```
torch.nn.utils.weight_norm.weight_norm(module,
name='weight', dim=0)[source]#

Parameters

Returns

Return type
```

Returns:

T_module

Code Examples

```
>>> m = weight_norm(nn.Linear(20, 40), name='weight')
>>> m
Linear(in_features=20, out_features=40, bias=True)
>>> m.weight_g.size()
torch.Size([40, 1])
>>> m.weight_v.size()
torch.Size([40, 20])
```

Notes & Warnings

⚠ WARNING

Warning This function is deprecated. Use torch.nn.utils.parametrizations.weight_norm() which uses the modern parametrization API. The new weight_norm is compatible with state_dict generated from old weight_norm. Migration guide: The magnitude (weight_g) and direction (weight_v) are now expressed as parametrizations.weight.original0 and parametrizations.weight.original1 respectively. If this is bothering you, please comment on pytorch/pytorch#102999 To remove the weight normalization reparametrization, use torch.nn.utils.parametrize.remove_parametrizations(). The weight is no longer recomputed once at module forward; instead, it will be recomputed on every access. To restore the old behavior, use torch.nn.utils.parametrize.cached() before invoking the module in question.

Detailed Documentation

Documentation

Apply weight normalization to a parameter in the given module.

Weight normalization is a reparameterization that decouples the magnitude of a weight tensor from its direction. This replaces the parameter specified by name (e.g. 'weight') with two parameters: one specifying the magnitude (e.g. 'weight_g') and one specifying the direction (e.g. 'weight_v'). Weight normalization is implemented via a hook that recomputes the weight tensor from the magnitude and direction before every forward() call.

By default, with dim=0, the norm is computed independently per output channel/plane. To compute a norm over the entire weight tensor, use dim=None.

See https://arxiv.org/abs/1602.07868

This function is deprecated. Use torch.nn.utils.parametrizations.weight_norm() which uses the modern parametrization API. The new weight_norm is compatible with state_dict generated from old weight_norm.

The magnitude (weight_g) and direction (weight_v) are now expressed as parametrizations.weight.original0 and parametrizations.weight.original1 respectively. If this is bothering you, please comment on pytorch/pytorch#102999

To remove the weight normalization reparametrization, use torch.nn.utils.parametrize.remove_parametrizations().

The weight is no longer recomputed once at module forward; instead, it will be recomputed on every access. To restore the old behavior, use torch.nn.utils.parametrize.cached() before invoking the module in question.

module (Module) – containing module

name (str, optional) – name of weight parameter

dim (int, optional) – dimension over which to compute the norm

The original module with the weight norm hook